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Helwan University

Faculty of Computer and Artificial Intelligence.

Course: Artificial Intelligence

Professor: Dr. Amr Ghoneim

Optical Character Recognition (OCR) Report

Team Members:

1. Ismael Mahmoud 20230083
2. Bavly Aziz 20230119
3. Habiba Khaled 20230171
4. Rahma Shaaban 20230202
5. Shahd Sayed 20230285
6. Kareem Ayman 20230413

Introduction:

I. Overview & Objectives:

The project focuses on building an **OCR** system for classifying English alphabet characters using the **EMNIST** dataset [1][7].

Two machine learning models, **Decision Trees** [4] and **Random Forests** [2], are compared when using them with **PCA** [2] and **HOG** [5] for feature extraction.

A **GUI** allows real-time predictions by uploading single-character images, and a **Jupyter Notebook** [6] documents the full experimental workflow.

Disclaimer: The data used in this model is already split, flattened, grayscale and in array format.

Note: there was a tiny problem in the test data that there were missing values in it but we solve it and we explain this point in experiments

II. Existing Applications:

Similar real-world applications projects that implement **OCR**:

- **Microsoft Lens**: Mobile/desktop tool for capturing photos of documents, whiteboards, or handwritten notes then converting the image to editable text.
- **Tesseract OCR:** A widely used open-source OCR engine. It accepts images (scans, photographs of documents, printed text, etc.) and extracts textual content.

Unlike deep learning-based approaches, this project focuses on classical machine learning techniques such as Decision Trees and Random Forests combined with handcrafted features (HOG and PCA). This allows better interpretability, lower computational cost, and clearer analysis of feature importance.

III. Literature Review:

Previous studies show that Decision Trees are simple and interpretable but often suffer from overfitting, especially with high-dimensional image features. This explains why, in our experiments, the Decision Tree achieved 65.9% accuracy with HOG using Gini Approach and dropped to 59.9% with PCA using Entropy approach, indicating limited generalization.

On the other hand, many works confirm that Random Forests improve performance by combining multiple decision trees and introducing randomness in both data sampling and feature selection. This reduces overfitting and leads to more stable predictions. Also, Random Forest achieved 87% accuracy using both HOG and PCA. Our results strongly agree with this, as Random Forest achieved 88.2% accuracy using HOG with Gini Approach and 82.0% using PCA with sqrt Approach, significantly outperforming the single Decision Tree.

Additionally, literature highlights that HOG features preserve local edge and shape information, which is crucial for handwritten character recognition. This aligns with our findings where HOG consistently outperformed PCA across both classifiers.

Overall, our experimental results are consistent with existing research, confirming that Random Forest + HOG is a more robust and reliable approach for handwritten letter classification compared to Decision Trees or PCA-based representation

Proposed Solution & Dataset:

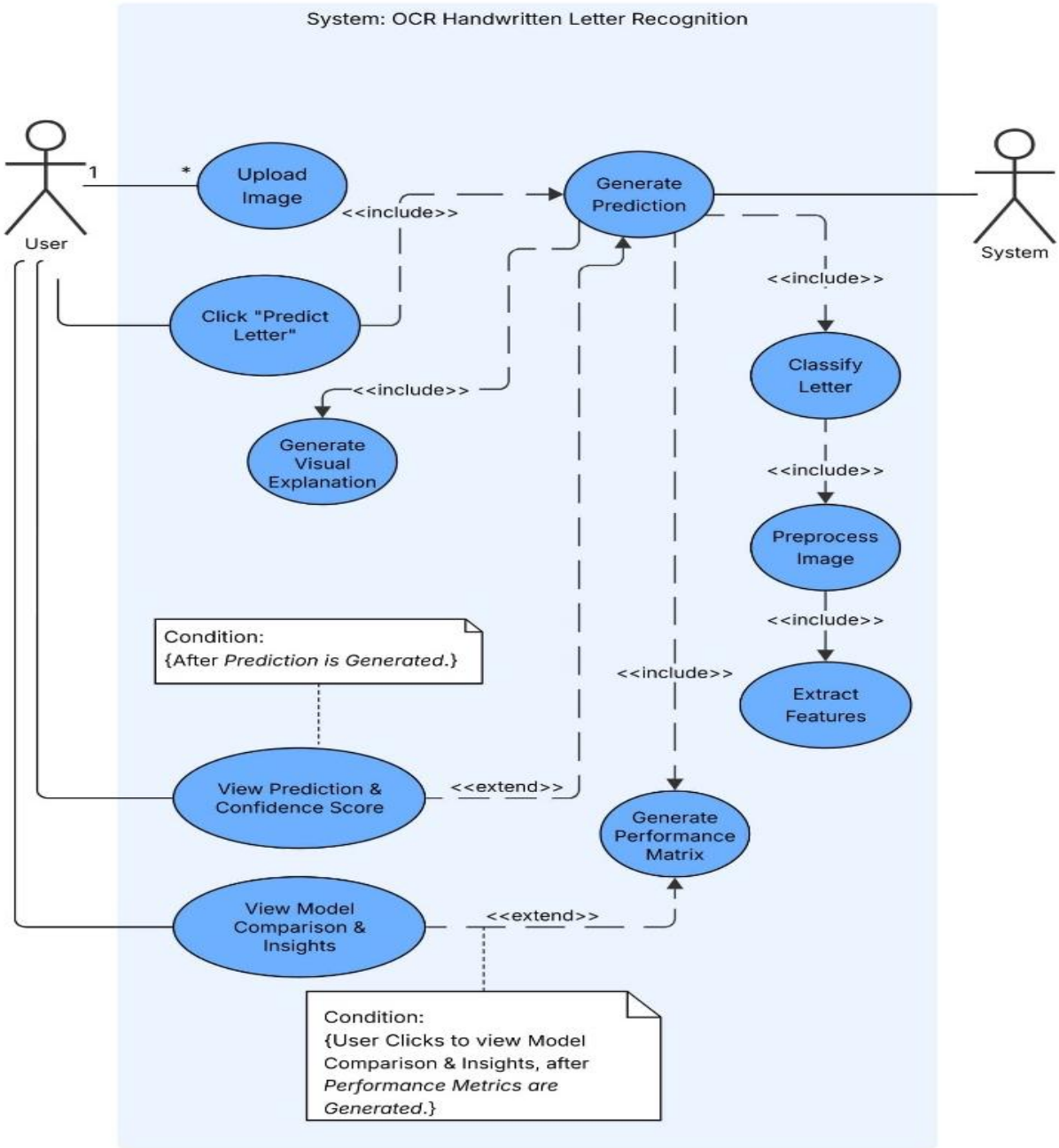
I. Key Functionalities from the User's Perspective:

This section describes the main functionalities of the proposed system from the users' perspective.

- **Image Upload**
  The user selects and uploads an image containing a handwritten letter. The system accepts the file and prepares it for processing.

- **Prediction**
  After uploading the image, the user clicks the **"Predict"** button. The system processes the image using the trained model and identifies the letter contained in the image.

- **Display Result**
  The system returns the predicted letter and displays it clearly to the user.



Refer to the **Use-case Diagram** below for further understanding of how users interact with our system (UI)

II. Dataset Employed:

The project uses the **EMNIST (Extended MNIST) dataset**, a publicly available dataset designed for handwritten character recognition. EMNIST extends the original MNIST dataset by including **handwritten English letters**, which makes it suitable for single-letter classification tasks. The dataset provides grayscale images of characters with labeled classes, enabling effective training and evaluation of machine learning models such as Decision Trees and Random Forests.

Applied Algorithms:

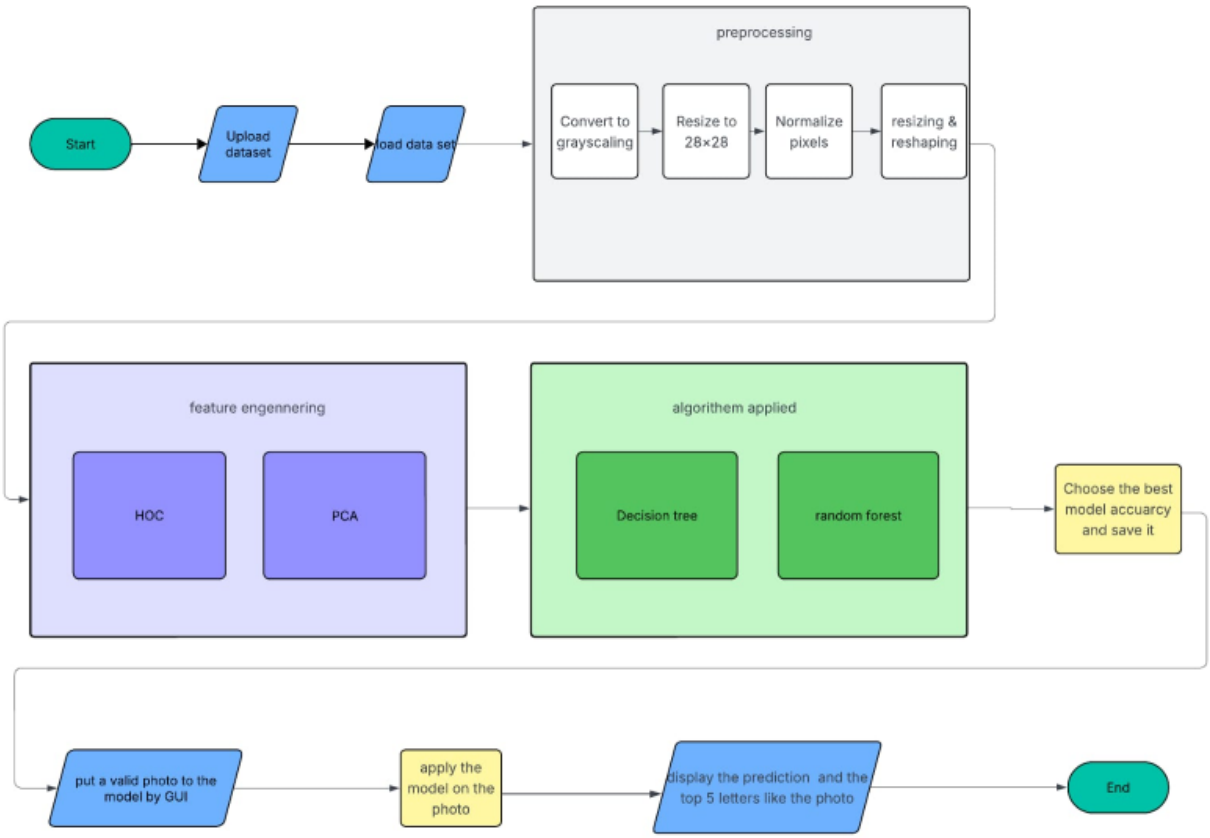
I. Development Environment & Tools:

This project was implemented in **Python**, using the following libraries:

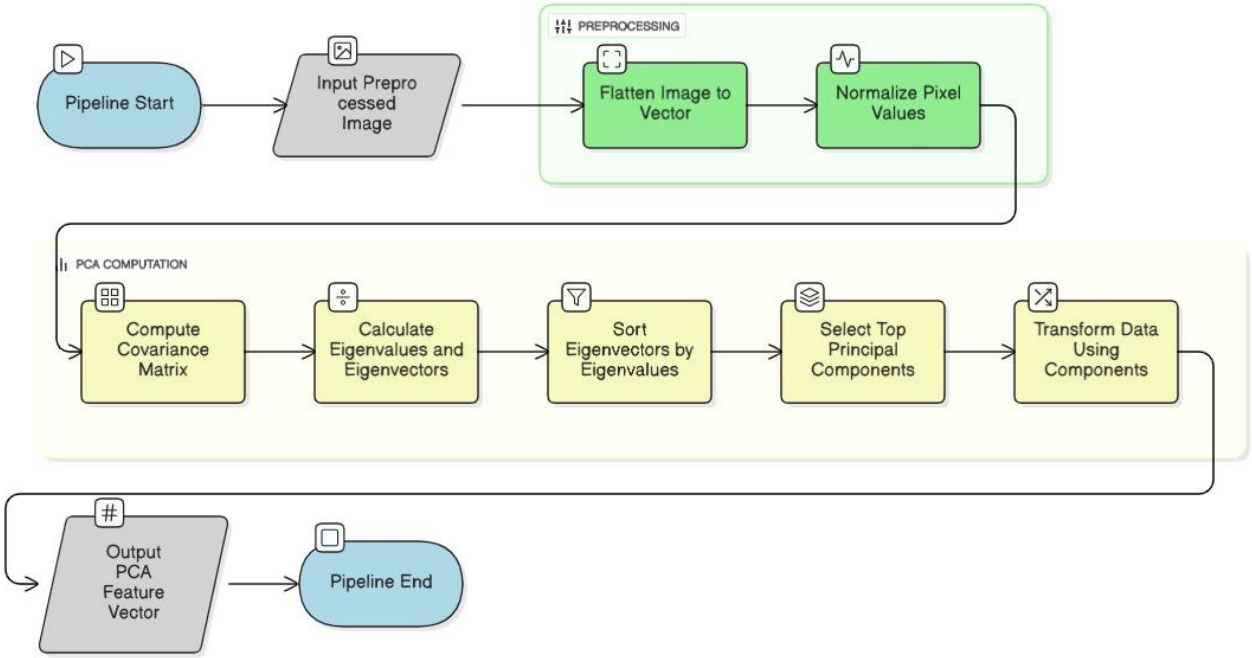
- **NumPy** and **Pandas**: for numerical operations and dataset handling.
- **Matplotlib** and **Seaborn**: for visualizing images, label distributions, and performance metrics.
- **scikit-learn**: for Decision Tree and Random Forest models, data splitting, scaling, and evaluation.
- **scikit-image (skimage.feature)**: for image preprocessing and HOG feature extraction.
- **Kagglehub (KaggleDatasetAdapter)**: to load EMNIST datasets directly from Kaggle.
- **Random**: for dataset shuffling and random initialization.
- **Joblib**: for saving and loading trained machine learning models efficiently and for parallel computation support during training or evaluation.

II. System Architecture & Block Diagrams:

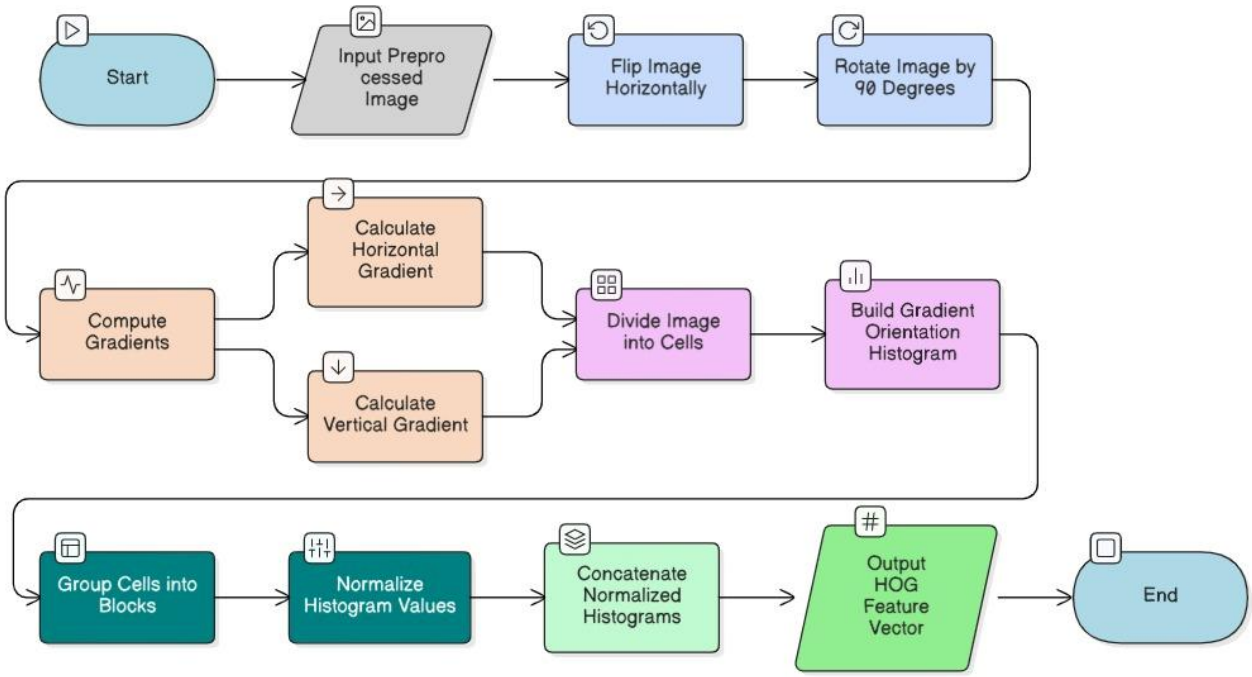
General System pipeline Flow Chart



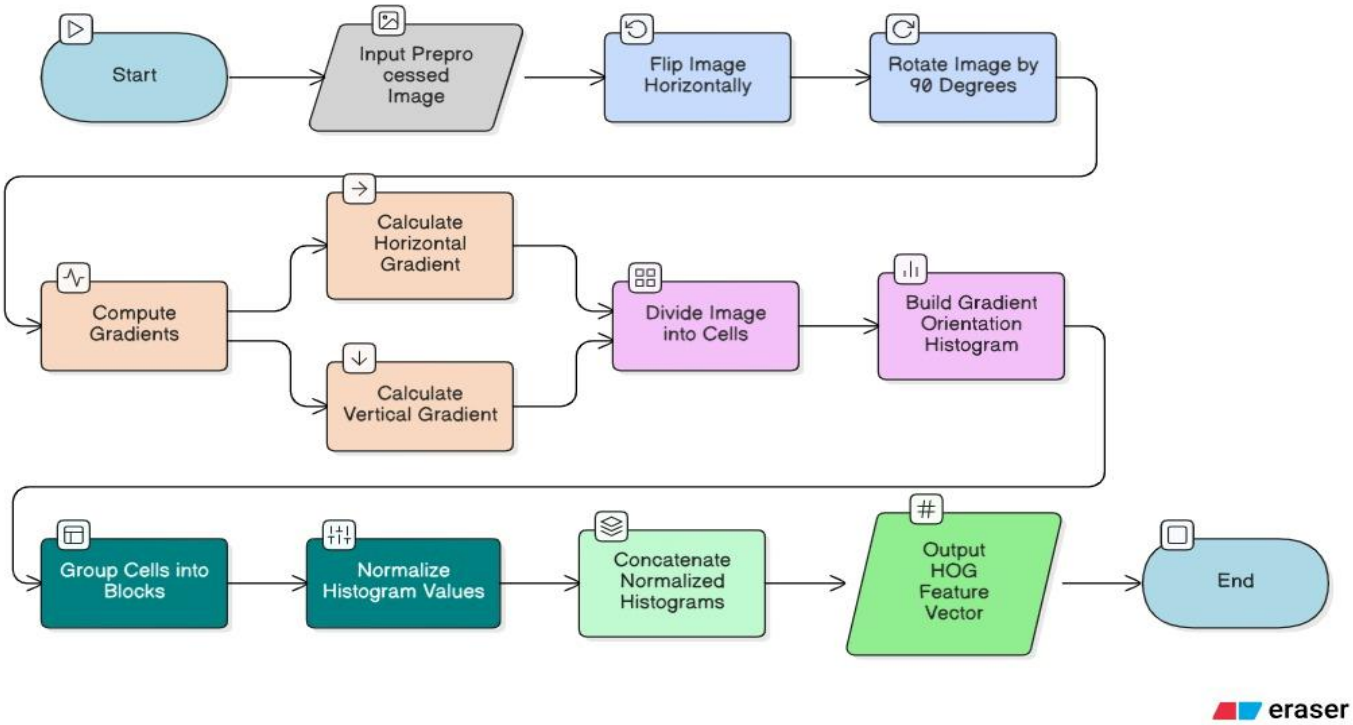
HOG Pipeline Flow Chart



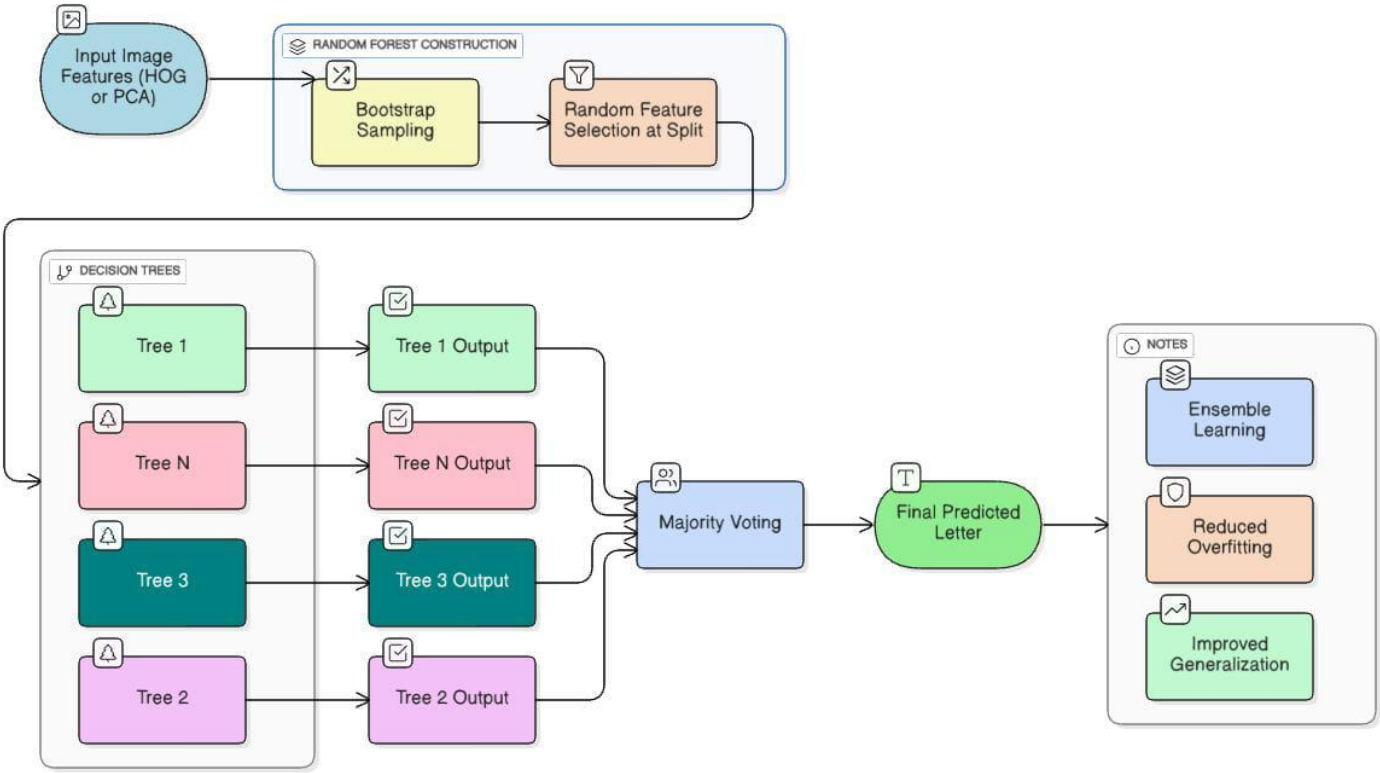
PCA Pipeline Flow Chart



Decision Tree Pipeline Flow Chart



Random Forest Pipeline Flow Chart



Experiments:

1. Dataset and Preprocessing

The EMNIST Letters dataset was used for training and evaluation.
Data preprocessing included:

- **Splitting** The dataset was originally divided into training and testing sets as provided; however, due to issues with the testing data, a portion of the training set was separated to create a validation set, ensuring the model was evaluated on all letters.
- **Normalization** of pixel values to the [0, 1]range.
- **Reshaping** the 784-pixel vectors back into **28×28 images**.
- **Fixing orientation** by applying a flip and 90° rotation to correct the EMNIST image format.
- **Exploratory plots**, including class distribution bar plots and sample visualizations, were used to analyze dataset balance and verify image correctness.

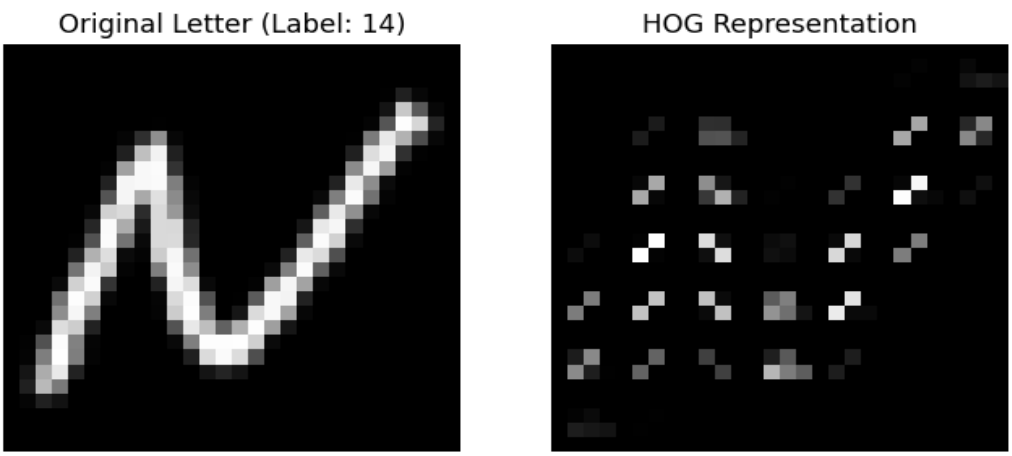
2. Feature Extraction Experiments

A. HOG (Histogram of Oriented Gradients)

HOG features were extracted using:

- 9 orientations
- 4×4 pixels per cell
- 2×2 cells per block

A sample HOG visualization was produced to show edge-based feature representation.



B. PCA (Principal Component Analysis)

PCA was applied to flattened pixel vectors for dimensionality reduction:

- Reduced from **784 features → 100 components**
- Explained-variance curve was plotted to confirm retained variance

3. Algorithms Used:

Decision Tree (DT):

- Trained on HOG features and on PCA-reduced features.
- Two split criteria were explicitly evaluated: **Gini index** and **Information Gain (Entropy)**, to assess the impact of impurity measures on classification performance.
- Model complexity was controlled using depth and minimum leaf size, with additional robustness assessed via 5-fold cross-validation.

Random Forest (RF):

- Evaluated on HOG features, PCA features, and a combined HOG + PCA representation.
- Tree randomization was introduced through feature subsampling using max_features, with both sqrt(n_features) and log2(n_features) considered.
- Ensemble size was fixed, and cross-validation was used to assess stability across folds.

4. Model and Testing Strategy

a. Cross-Validation

All models were evaluated using **5-fold cross-validation** on the training set to assess general performance stability.

b. Classifiers Tested

Two classifiers were applied to both HOG and PCA feature sets:

I. **Decision Tree Classifier**

- Hyperparameters tested: max_depth=20, min_samples_leaf=5
- Evaluated using CV accuracy per fold, final test accuracy, classification report, and confusion matrix

II. **Random Forest Classifier**

- 100 decision trees (n_estimators=100)
- Used full parallelization (n_jobs=-1)
- Evaluated using the same metrics as above

Results

I. Using PCA vs HOG:

After running tests using both **PCA** and **HOG,** it was found that using **HOG** method for feature extraction renders better results, which was to be expected for this type of dataset. This is because of the different ways each method handles the feature extraction.

For most image-based machine learning tasks, **HOG provides better accuracy** because it extracts **meaningful shape and edge information** (refer to figure 1), whereas PCA mainly compresses pixel data without regard to visual structure.

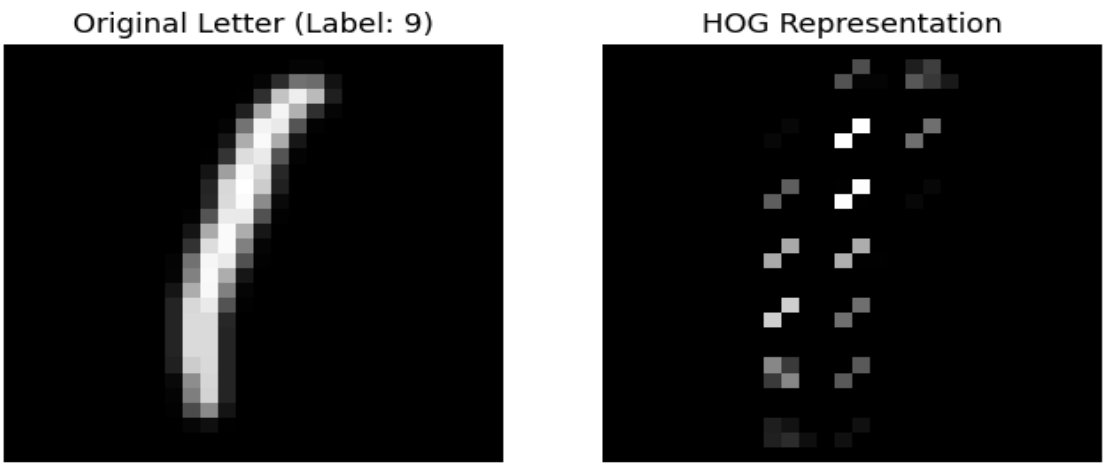


Figure 1

II. Using Decision Trees vs Random Forests

Decision Tree:

Decision Tree performance was notably affected by both the split criterion and the feature representation. Overall trends observed are summarized below:

- **Entropy vs. Gini:** Models using the ***Entropy*** **criterion achieved higher accuracy** and more balanced class predictions than those using the *Gini index*, indicating superior impurity reduction.

- These results confirm that single-tree models are sensitive to both impurity measures and feature dimensionality.

Random Forest:

Random Forest models consistently outperformed Decision Trees due to ensemble averaging and inherent tree randomisation. The following observations highlight the impact of randomisation and feature handling:

- **Feature Subsampling:** Using max_features = sqrt(n_features) produced higher accuracy than log2(n_features), as it preserved stronger split decisions while maintaining sufficient randomness.
- **Effect of PCA:** PCA-based Random Forests showed improved stability and efficiency compared to raw HOG features.
- **Combined Representation:** The HOG + PCA configuration delivered the most robust and balanced performance, benefiting from both rich feature extraction and dimensionality reduction.

Overall, controlled randomisation and appropriate feature compression were key factors in achieving superior accuracy and generalization.

Methods & Algorithms	Accuracy
Decision Tree HOG Accuracy on validation data	**65.91%**
Decision Tree HOG Accuracy on test data Gini approach	**64.54%**
Decision Tree HOG Accuracy on validation data Entropy Approach	**67.90%**
Decision Tree HOG Accuracy on test data Entropy approach	**66.03%**
Decision Tree PCA on validate data Accuracy Entropy approach	**59.92%**
Decision Tree PCA on the test data Accuracy Entropy approach	**58.67%**
Random Forest HOG Accuracy on the validation data log2 approach	**88.54%**
Random Forest HOG Accuracy on the test data log2 approach	**87.07%**
Random Forest HOG Accuracy on the validation data sqrt approach	**88.87%**
Random Forest HOG Accuracy on the test data sqrt approach	**87.95%**
Random Forest PCA Accuracy on the validation data sqrt approach	**83.09%**
Random Forest PCA Accuracy on the test data sqrt approach	**81.26%**
Random Forest with HOG + PCA Accuracy on test data	**86.30%**

I. Hog + Decision tree class-wise accuracy before using validation

Decision Tree Accuracy on test data: 0.6454

Classification Report on the test data:

	precision	recall	f1-score	support
1	0.54	0.51	0.53	799
2	0.73	0.71	0.72	800
3	0.75	0.73	0.74	800
4	0.72	0.65	0.69	800
5	0.73	0.72	0.72	800
6	0.75	0.69	0.72	800
7	0.50	0.44	0.47	800
8	0.72	0.67	0.69	800
9	0.56	0.53	0.55	800
10	0.81	0.72	0.76	800
11	0.74	0.65	0.69	800
12	0.57	0.54	0.55	800
13	0.79	0.72	0.75	800
14	0.72	0.57	0.64	800
15	0.78	0.75	0.77	800
16	0.81	0.81	0.81	800
17	0.55	0.48	0.52	800
18	0.72	0.62	0.67	800
19	0.71	0.83	0.77	400
20	0.00	0.00	0.00	0
21	0.00	0.00	0.00	0
22	0.00	0.00	0.00	0
23	0.00	0.00	0.00	0
24	0.00	0.00	0.00	0
25	0.00	0.00	0.00	0
26	0.00	0.00	0.00	0
accuracy			0.65	14799
macro avg	0.51	0.48	0.49	14799
weighted avg	0.69	0.65	0.67	14799

II. Hog + Decision tree class-wise accuracy After using validation

Decision Tree Accuracy on validation data: 0.6591

Classification Report on the validation data:

	precision	recall	f1-score	support
1	0.49	0.47	0.48	679
2	0.66	0.70	0.68	679
3	0.72	0.73	0.72	684
4	0.68	0.67	0.67	680
5	0.67	0.68	0.67	688
6	0.68	0.67	0.68	679
7	0.43	0.44	0.43	677
8	0.67	0.69	0.68	685
9	0.53	0.56	0.54	686
10	0.72	0.71	0.71	680
11	0.61	0.58	0.59	688
12	0.57	0.54	0.55	683
13	0.72	0.73	0.73	680
14	0.62	0.62	0.62	673
15	0.79	0.76	0.77	682
16	0.76	0.77	0.76	686
17	0.50	0.51	0.51	687
18	0.62	0.63	0.63	684
19	0.78	0.79	0.79	678
20	0.66	0.68	0.67	687
21	0.67	0.66	0.66	684
22	0.71	0.73	0.72	684
23	0.71	0.68	0.70	684
24	0.71	0.66	0.69	687
25	0.67	0.68	0.68	691
26	0.82	0.80	0.81	685
accuracy			0.66	17760
macro avg	0.66	0.66	0.66	17760
weighted avg	0.66	0.66	0.66	17760

III. PCA + Decision Tree class-wise accuracy before using validation

```
Decision Tree PCA on the test data Accuracy: 0.5762

Classification Report:
              precision    recall  f1-score   support

           1       0.50      0.44      0.47       799
           2       0.62      0.55      0.58       800
           3       0.67      0.68      0.68       800
           4       0.63      0.56      0.60       800
           5       0.68      0.61      0.64       800
           6       0.63      0.55      0.59       800
           7       0.46      0.38      0.41       800
           8       0.65      0.59      0.62       800
           9       0.58      0.56      0.57       800
          10       0.77      0.67      0.72       800
          11       0.66      0.56      0.60       800
          12       0.58      0.55      0.56       800
          13       0.83      0.76      0.79       800
          14       0.64      0.54      0.59       800
          15       0.71      0.71      0.71       800
          16       0.72      0.67      0.69       800
          17       0.51      0.42      0.46       800
          18       0.64      0.54      0.58       800
          19       0.53      0.66      0.59       400
          20       0.00      0.00      0.00         0
          21       0.00      0.00      0.00         0
          22       0.00      0.00      0.00         0
          23       0.00      0.00      0.00         0
          24       0.00      0.00      0.00         0
          25       0.00      0.00      0.00         0
          26       0.00      0.00      0.00         0

    accuracy                           0.58     14799
   macro avg       0.46      0.42      0.44     14799
weighted avg       0.63      0.58      0.60     14799
```

IV. PCA + Decision Tree class-wise accuracy After using validation

Decision Tree PCA on validate data Accuracy: 0.5988

Classification Report:

	precision	recall	f1-score	support
1	0.48	0.45	0.46	679
2	0.58	0.56	0.57	679
3	0.65	0.67	0.66	684
4	0.53	0.54	0.53	680
5	0.61	0.59	0.60	688
6	0.59	0.57	0.58	679
7	0.37	0.36	0.36	677
8	0.61	0.62	0.61	685
9	0.50	0.54	0.52	686
10	0.67	0.65	0.66	680
11	0.52	0.50	0.51	688
12	0.53	0.54	0.53	683
13	0.77	0.79	0.78	680
14	0.60	0.57	0.59	673
15	0.70	0.74	0.72	682
16	0.65	0.67	0.66	686
17	0.44	0.44	0.44	687
18	0.53	0.55	0.54	684
19	0.66	0.65	0.66	678
20	0.61	0.60	0.60	687
21	0.61	0.60	0.61	684
22	0.65	0.71	0.68	684
23	0.72	0.68	0.70	684
24	0.67	0.68	0.67	687
25	0.63	0.62	0.63	691
26	0.67	0.69	0.68	685
accuracy			0.60	17760
macro avg	0.60	0.60	0.60	17760
weighted avg	0.60	0.60	0.60	17760

V. HOG + Random Forests class-wise accuracy before using validation

Random Forest Accuracy on the test data: 0.8795

Classification Report:

	precision	recall	f1-score	support
1	0.83	0.89	0.86	799
2	0.92	0.92	0.92	800
3	0.91	0.91	0.91	800
4	0.92	0.90	0.91	800
5	0.90	0.92	0.91	800
6	0.96	0.92	0.94	800
7	0.87	0.67	0.76	800
8	0.91	0.90	0.91	800
9	0.72	0.76	0.74	800
10	0.95	0.90	0.92	800
11	0.93	0.93	0.93	800
12	0.76	0.73	0.74	800
13	0.96	0.97	0.96	800
14	0.94	0.89	0.92	800
15	0.89	0.97	0.93	800
16	0.96	0.96	0.96	800
17	0.78	0.81	0.79	800
18	0.95	0.85	0.90	800
19	0.93	0.95	0.94	400
20	0.00	0.00	0.00	0
21	0.00	0.00	0.00	0
22	0.00	0.00	0.00	0
23	0.00	0.00	0.00	0
24	0.00	0.00	0.00	0
25	0.00	0.00	0.00	0
26	0.00	0.00	0.00	0
accuracy			0.88	14799
macro avg	0.65	0.64	0.65	14799
weighted avg	0.89	0.88	0.89	14799

VI. HOG + Random Forests class-wise accuracy After using validation

Random Forest Accuracy on the validation data: 0.8887

Classification Report:

	precision	recall	f1-score	support
1	0.82	0.86	0.84	679
2	0.89	0.92	0.90	679
3	0.91	0.92	0.92	684
4	0.89	0.91	0.90	680
5	0.89	0.90	0.90	688
6	0.92	0.93	0.93	679
7	0.87	0.67	0.76	677
8	0.89	0.91	0.90	685
9	0.72	0.76	0.74	686
10	0.90	0.90	0.90	680
11	0.91	0.92	0.91	688
12	0.77	0.71	0.74	683
13	0.94	0.94	0.94	680
14	0.89	0.89	0.89	673
15	0.90	0.96	0.93	682
16	0.93	0.93	0.93	686
17	0.77	0.82	0.79	687
18	0.89	0.85	0.87	684
19	0.95	0.95	0.95	678
20	0.90	0.92	0.91	687
21	0.91	0.93	0.92	684
22	0.91	0.91	0.91	684
23	0.94	0.94	0.94	684
24	0.96	0.92	0.94	687
25	0.92	0.88	0.90	691
26	0.93	0.96	0.94	685
accuracy			0.89	17760
macro avg	0.89	0.89	0.89	17760
weighted avg	0.89	0.89	0.89	17760

VII. PCA + Random Forest classes-wise accuracy before using validation

Random Forest PCA Accuracy on the test data: 0.8126

Classification Report:

	precision	recall	f1-score	support
1	0.77	0.78	0.77	799
2	0.86	0.84	0.85	800
3	0.87	0.91	0.89	800
4	0.86	0.78	0.82	800
5	0.87	0.86	0.86	800
6	0.86	0.83	0.85	800
7	0.81	0.60	0.69	800
8	0.84	0.82	0.83	800
9	0.72	0.70	0.71	800
10	0.87	0.87	0.87	800
11	0.86	0.83	0.85	800
12	0.75	0.71	0.73	800
13	0.90	0.93	0.92	800
14	0.85	0.79	0.82	800
15	0.86	0.94	0.90	800
16	0.85	0.91	0.88	800
17	0.75	0.70	0.72	800
18	0.92	0.75	0.83	800
19	0.87	0.93	0.89	400
20	0.00	0.00	0.00	0
21	0.00	0.00	0.00	0
22	0.00	0.00	0.00	0
23	0.00	0.00	0.00	0
24	0.00	0.00	0.00	0
25	0.00	0.00	0.00	0
26	0.00	0.00	0.00	0
accuracy			0.81	14799
macro avg	0.61	0.60	0.60	14799
weighted avg	0.84	0.81	0.82	14799

VIII. PCA + Random Forest classes-wise accuracy After using validation

Random Forest PCA Accuracy on the validation data: 0.8309

Classification Report:

	precision	recall	f1-score	support
1	0.74	0.75	0.74	679
2	0.84	0.80	0.82	679
3	0.85	0.92	0.88	684
4	0.85	0.81	0.83	680
5	0.84	0.85	0.84	688
6	0.82	0.84	0.83	679
7	0.77	0.61	0.68	677
8	0.83	0.84	0.83	685
9	0.72	0.72	0.72	686
10	0.81	0.85	0.83	680
11	0.83	0.81	0.82	688
12	0.74	0.71	0.72	683
13	0.86	0.93	0.89	680
14	0.84	0.84	0.84	673
15	0.84	0.93	0.89	682
16	0.84	0.88	0.86	686
17	0.72	0.74	0.73	687
18	0.85	0.75	0.80	684
19	0.92	0.92	0.92	678
20	0.84	0.85	0.85	687
21	0.84	0.88	0.86	684
22	0.88	0.88	0.88	684
23	0.89	0.90	0.90	684
24	0.88	0.87	0.87	687
25	0.85	0.80	0.83	691
26	0.90	0.92	0.91	685
accuracy			0.83	17760
macro avg	0.83	0.83	0.83	17760
weighted avg	0.83	0.83	0.83	17760

IX. PCA & HOG Random Forest classes-wise accuracy before using validation

Random Forest with HOG + PCA Accuracy on test data: 0.8636

Classification Report on test data:

	precision	recall	f1-score	support
1	0.83	0.86	0.84	799
2	0.91	0.90	0.91	800
3	0.87	0.91	0.89	800
4	0.93	0.89	0.91	800
5	0.90	0.90	0.90	800
6	0.93	0.89	0.91	800
7	0.86	0.67	0.75	800
8	0.90	0.86	0.88	800
9	0.72	0.72	0.72	800
10	0.94	0.89	0.91	800
11	0.91	0.91	0.91	800
12	0.73	0.73	0.73	800
13	0.94	0.94	0.94	800
14	0.90	0.87	0.88	800
15	0.89	0.96	0.92	800
16	0.94	0.94	0.94	800
17	0.80	0.81	0.81	800
18	0.94	0.85	0.89	800
19	0.92	0.96	0.94	400
20	0.00	0.00	0.00	0
21	0.00	0.00	0.00	0
22	0.00	0.00	0.00	0
23	0.00	0.00	0.00	0
24	0.00	0.00	0.00	0
25	0.00	0.00	0.00	0
26	0.00	0.00	0.00	0
accuracy			0.86	14799
macro avg	0.64	0.63	0.64	14799
weighted avg	0.88	0.86	0.87	14799

X. PCA & HOG Random Forest classes-wise accuracy
After using validation

```
Random Forest with HOG + PCA Accuracy on validation data: 0.8776

Classification Report on validation data:
              precision    recall  f1-score   support

           1       0.81      0.85      0.83       679
           2       0.90      0.90      0.90       679
           3       0.88      0.93      0.91       684
           4       0.88      0.92      0.90       680
           5       0.89      0.89      0.89       688
           6       0.90      0.89      0.90       679
           7       0.83      0.66      0.74       677
           8       0.88      0.88      0.88       685
           9       0.72      0.73      0.73       686
          10       0.89      0.87      0.88       680
          11       0.88      0.91      0.90       688
          12       0.75      0.71      0.73       683
          13       0.92      0.93      0.93       680
          14       0.88      0.88      0.88       673
          15       0.90      0.97      0.93       682
          16       0.90      0.93      0.92       686
          17       0.77      0.79      0.78       687
          18       0.90      0.85      0.88       684
          19       0.94      0.96      0.95       678
          20       0.86      0.90      0.88       687
          21       0.90      0.93      0.91       684
          22       0.90      0.89      0.90       684
          23       0.94      0.92      0.93       684
          24       0.95      0.92      0.93       687
          25       0.91      0.85      0.88       691
          26       0.93      0.95      0.94       685

    accuracy                           0.88     17760
   macro avg       0.88      0.88      0.88     17760
weighted avg       0.88      0.88      0.88     17760
```

XI. Decision Tree + HOG Validation Data GINI Approach Accuracy Class-wise

```
Making predictions on validation data Gini Approach...
Decision Tree Accuracy on validation data: 0.6591

Classification Report on the validation data:
              precision    recall  f1-score   support

           1       0.49      0.47      0.48       679
           2       0.66      0.70      0.68       679
           3       0.72      0.73      0.72       684
           4       0.68      0.67      0.67       680
           5       0.67      0.68      0.67       688
           6       0.68      0.67      0.68       679
           7       0.43      0.44      0.43       677
           8       0.67      0.69      0.68       685
           9       0.53      0.56      0.54       686
          10       0.72      0.71      0.71       680
          11       0.61      0.58      0.59       688
          12       0.57      0.54      0.55       683
          13       0.72      0.73      0.73       680
          14       0.62      0.62      0.62       673
          15       0.79      0.76      0.77       682
          16       0.76      0.77      0.76       686
          17       0.50      0.51      0.51       687
          18       0.62      0.63      0.63       684
          19       0.78      0.79      0.79       678
          20       0.66      0.68      0.67       687
          21       0.67      0.66      0.66       684
          22       0.71      0.73      0.72       684
          23       0.71      0.68      0.70       684
          24       0.71      0.66      0.69       687
          25       0.67      0.68      0.68       691
          26       0.82      0.80      0.81       685

    accuracy                           0.66     17760
   macro avg       0.66      0.66      0.66     17760
weighted avg       0.66      0.66      0.66     17760
```

XII. Decision Tree + HOG Test Data GINI Approach Accuracy Class-wise

```
Decision Tree Accuracy on test data gini approach: 0.6454

Classification Report on the test data:
              precision    recall  f1-score   support

           1       0.54      0.51      0.53       799
           2       0.73      0.71      0.72       800
           3       0.75      0.73      0.74       800
           4       0.72      0.65      0.69       800
           5       0.73      0.72      0.72       800
           6       0.75      0.69      0.72       800
           7       0.50      0.44      0.47       800
           8       0.72      0.67      0.69       800
           9       0.56      0.53      0.55       800
          10       0.81      0.72      0.76       800
          11       0.74      0.65      0.69       800
          12       0.57      0.54      0.55       800
          13       0.79      0.72      0.75       800
          14       0.72      0.57      0.64       800
          15       0.78      0.75      0.77       800
          16       0.81      0.81      0.81       800
          17       0.55      0.48      0.52       800
          18       0.72      0.62      0.67       800
          19       0.71      0.83      0.77       400
          20       0.00      0.00      0.00         0
          21       0.00      0.00      0.00         0
          22       0.00      0.00      0.00         0
          23       0.00      0.00      0.00         0
          24       0.00      0.00      0.00         0
          25       0.00      0.00      0.00         0
          26       0.00      0.00      0.00         0

    accuracy                           0.65     14799
   macro avg       0.51      0.48      0.49     14799
weighted avg       0.69      0.65      0.67     14799
```

XIII. Decision Tree + HOG Test Data Entropy Approach Accuracy Class-wise

```
Decision Tree Accuracy on test data Entropy approach: 0.6603

Classification Report on the test data:
              precision    recall  f1-score   support

           1       0.59      0.53      0.56       799
           2       0.70      0.68      0.69       800
           3       0.76      0.74      0.75       800
           4       0.76      0.68      0.71       800
           5       0.72      0.67      0.69       800
           6       0.79      0.70      0.74       800
           7       0.52      0.47      0.50       800
           8       0.75      0.66      0.70       800
           9       0.60      0.58      0.59       800
          10       0.82      0.75      0.78       800
          11       0.76      0.69      0.72       800
          12       0.62      0.59      0.61       800
          13       0.81      0.74      0.77       800
          14       0.72      0.63      0.67       800
          15       0.76      0.77      0.77       800
          16       0.80      0.78      0.79       800
          17       0.55      0.49      0.52       800
          18       0.71      0.67      0.69       800
          19       0.72      0.80      0.76       400
          20       0.00      0.00      0.00         0
          21       0.00      0.00      0.00         0
          22       0.00      0.00      0.00         0
          23       0.00      0.00      0.00         0
          24       0.00      0.00      0.00         0
          25       0.00      0.00      0.00         0
          26       0.00      0.00      0.00         0

    accuracy                           0.66     14799
   macro avg       0.52      0.49      0.50     14799
weighted avg       0.71      0.66      0.68     14799
```

XIV. Decision Tree + HOG Validation Data Entropy Approach Accuracy Class-wise

```
Making predictions on validation data entropy Approach...
Decision Tree Accuracy on validation data: 0.6790

Classification Report on the validation data:
              precision    recall  f1-score   support

           1       0.53      0.53      0.53       679
           2       0.65      0.67      0.66       679
           3       0.75      0.76      0.75       684
           4       0.66      0.69      0.68       680
           5       0.68      0.67      0.68       688
           6       0.71      0.71      0.71       679
           7       0.47      0.50      0.48       677
           8       0.70      0.68      0.69       685
           9       0.56      0.59      0.57       686
          10       0.70      0.73      0.71       680
          11       0.67      0.66      0.66       688
          12       0.59      0.59      0.59       683
          13       0.76      0.76      0.76       680
          14       0.66      0.65      0.65       673
          15       0.77      0.76      0.76       682
          16       0.76      0.74      0.75       686
          17       0.53      0.50      0.52       687
          18       0.63      0.64      0.64       684
          19       0.81      0.77      0.79       678
          20       0.72      0.72      0.72       687
          21       0.69      0.68      0.69       684
          22       0.75      0.74      0.75       684
          23       0.72      0.72      0.72       684
          24       0.71      0.70      0.70       687
          25       0.70      0.70      0.70       691
          26       0.81      0.80      0.80       685

    accuracy                           0.68     17760
   macro avg       0.68      0.68      0.68     17760
weighted avg       0.68      0.68      0.68     17760
```

XV. Random Forest + HOG Validation Data Log2 Approach Accuracy Class-wise

```
Making predictions on the validation data log2 approach...
Random Forest Accuracy on the validation data log2 approach: 0.8854

Classification Report:
              precision    recall  f1-score   support

           1       0.84      0.84      0.84       679
           2       0.89      0.91      0.90       679
           3       0.91      0.93      0.92       684
           4       0.90      0.90      0.90       680
           5       0.90      0.90      0.90       688
           6       0.92      0.94      0.93       679
           7       0.86      0.66      0.75       677
           8       0.88      0.90      0.89       685
           9       0.73      0.74      0.73       686
          10       0.90      0.89      0.89       680
          11       0.90      0.92      0.91       688
          12       0.75      0.72      0.74       683
          13       0.91      0.95      0.93       680
          14       0.89      0.88      0.88       673
          15       0.88      0.96      0.92       682
          16       0.94      0.94      0.94       686
          17       0.77      0.83      0.80       687
          18       0.90      0.84      0.87       684
          19       0.94      0.95      0.95       678
          20       0.88      0.91      0.90       687
          21       0.90      0.92      0.91       684
          22       0.92      0.90      0.91       684
          23       0.94      0.93      0.93       684
          24       0.96      0.92      0.94       687
          25       0.91      0.88      0.89       691
          26       0.92      0.96      0.94       685

    accuracy                           0.89     17760
   macro avg       0.89      0.89      0.88     17760
weighted avg       0.89      0.89      0.88     17760
```

XVI. Random Forest + HOG Test Data Log2 Approach Accuracy Class-wise

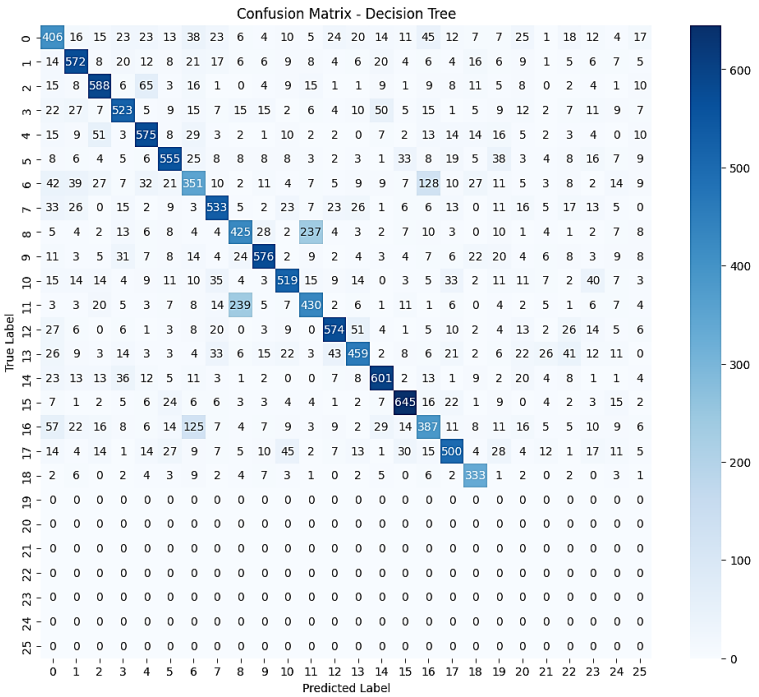
```
Making predictions on the test data log2 approach...
Random Forest Accuracy on the test data log2 approach: 0.8707

Classification Report:
              precision    recall  f1-score   support

           1       0.83      0.85      0.84       799
           2       0.91      0.92      0.91       800
           3       0.92      0.91      0.92       800
           4       0.92      0.88      0.90       800
           5       0.90      0.93      0.91       800
           6       0.95      0.92      0.93       800
           7       0.87      0.63      0.73       800
           8       0.89      0.90      0.90       800
           9       0.72      0.74      0.73       800
          10       0.95      0.90      0.92       800
          11       0.91      0.91      0.91       800
          12       0.75      0.74      0.75       800
          13       0.94      0.97      0.96       800
          14       0.93      0.87      0.90       800
          15       0.87      0.97      0.92       800
          16       0.94      0.96      0.95       800
          17       0.78      0.80      0.79       800
          18       0.95      0.83      0.89       800
          19       0.91      0.96      0.93       400
          20       0.00      0.00      0.00         0
          21       0.00      0.00      0.00         0
          22       0.00      0.00      0.00         0
          23       0.00      0.00      0.00         0
          24       0.00      0.00      0.00         0
          25       0.00      0.00      0.00         0
          26       0.00      0.00      0.00         0

    accuracy                           0.87     14799
   macro avg       0.65      0.64      0.64     14799
weighted avg       0.89      0.87      0.88     14799
```

XVLL. Random Forest + HOG Validation Data Sqrt Approach Accuracy Class-wise (BEST MODEL)

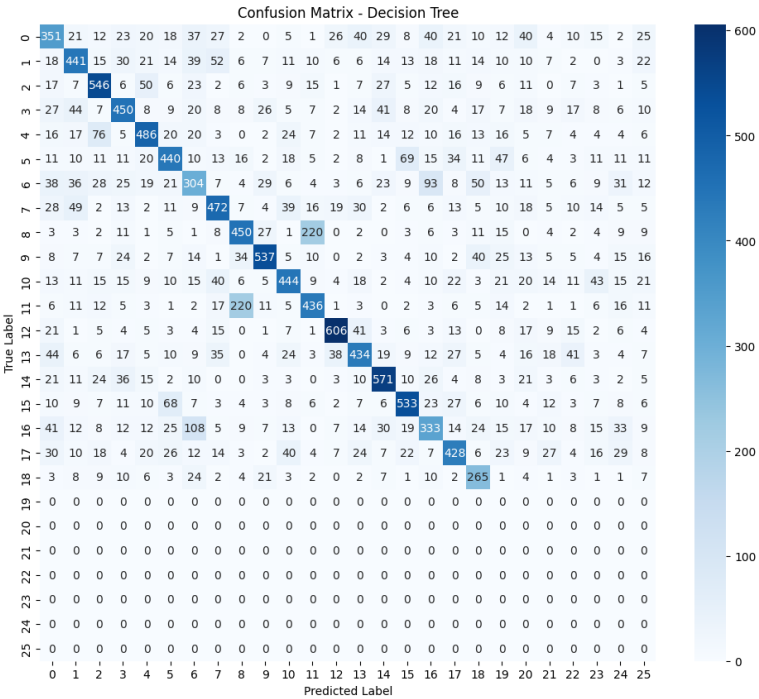
```
Making predictions on the validation data...
Random Forest Accuracy on the validation data: 0.8887

Classification Report:
              precision    recall  f1-score   support

           1       0.82      0.86      0.84       679
           2       0.89      0.92      0.90       679
           3       0.91      0.92      0.92       684
           4       0.89      0.91      0.90       680
           5       0.89      0.90      0.90       688
           6       0.92      0.93      0.93       679
           7       0.87      0.67      0.76       677
           8       0.89      0.91      0.90       685
           9       0.72      0.76      0.74       686
          10       0.90      0.90      0.90       680
          11       0.91      0.92      0.91       688
          12       0.77      0.71      0.74       683
          13       0.94      0.94      0.94       680
          14       0.89      0.89      0.89       673
          15       0.90      0.96      0.93       682
          16       0.93      0.93      0.93       686
          17       0.77      0.82      0.79       687
          18       0.89      0.85      0.87       684
          19       0.95      0.95      0.95       678
          20       0.90      0.92      0.91       687
          21       0.91      0.93      0.92       684
          22       0.91      0.91      0.91       684
          23       0.94      0.94      0.94       684
          24       0.96      0.92      0.94       687
          25       0.92      0.88      0.90       691
          26       0.93      0.96      0.94       685

    accuracy                           0.89     17760
   macro avg       0.89      0.89      0.89     17760
weighted avg       0.89      0.89      0.89     17760
```

Before creating a validation set, ensuring the model was evaluated on all letters.

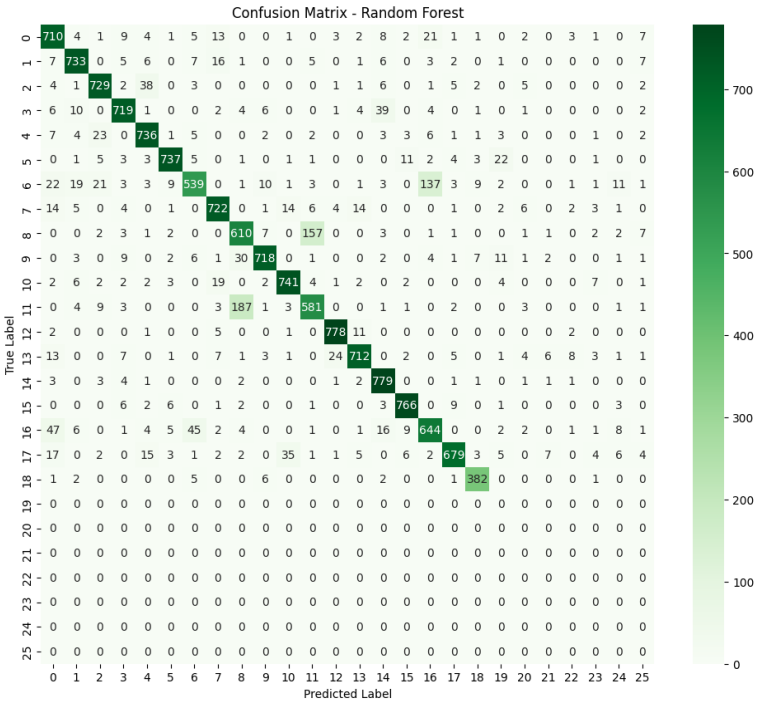
HOG & Decision Tree



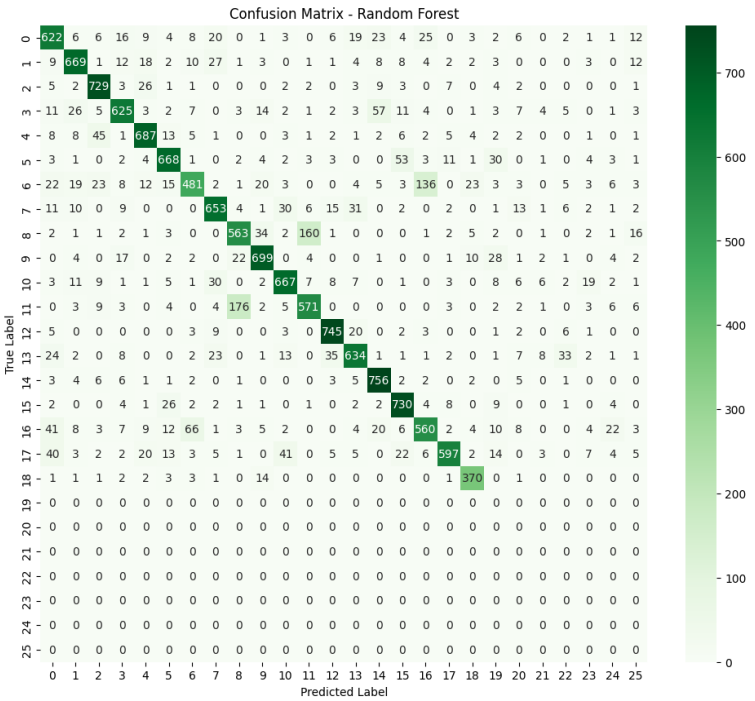
PCA & Decision Tree



HOG & Random Forests

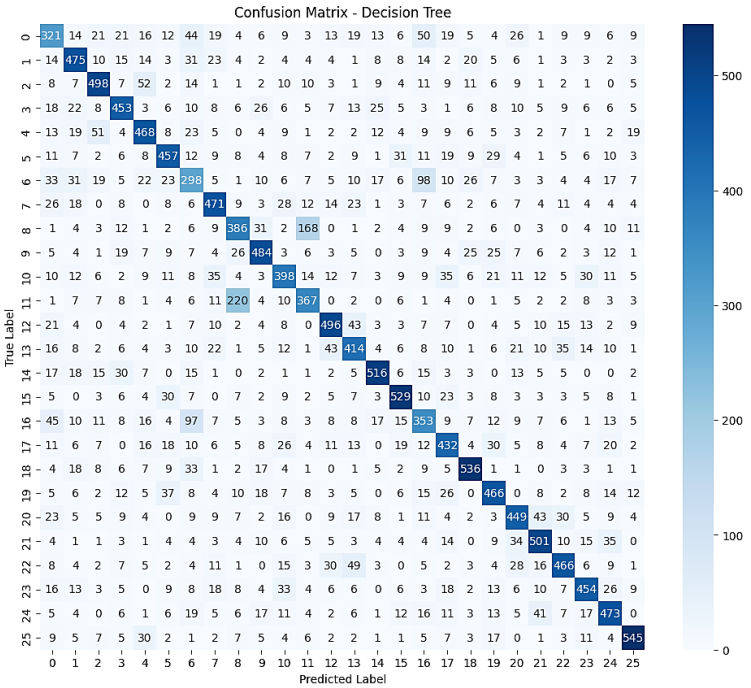


PCA & Random Forests

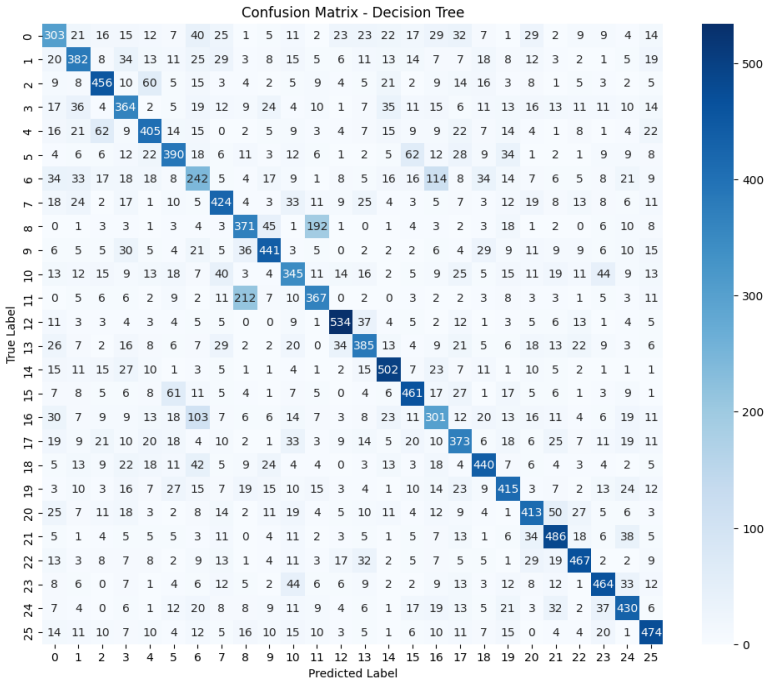


After creating a validation set, ensuring the model was evaluated on all letters.

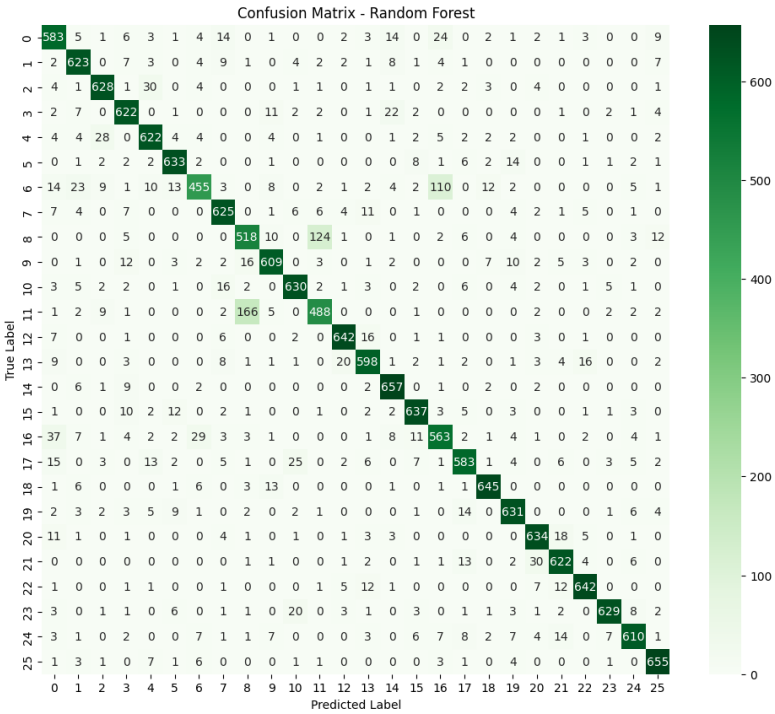
HOG & Decision Tree



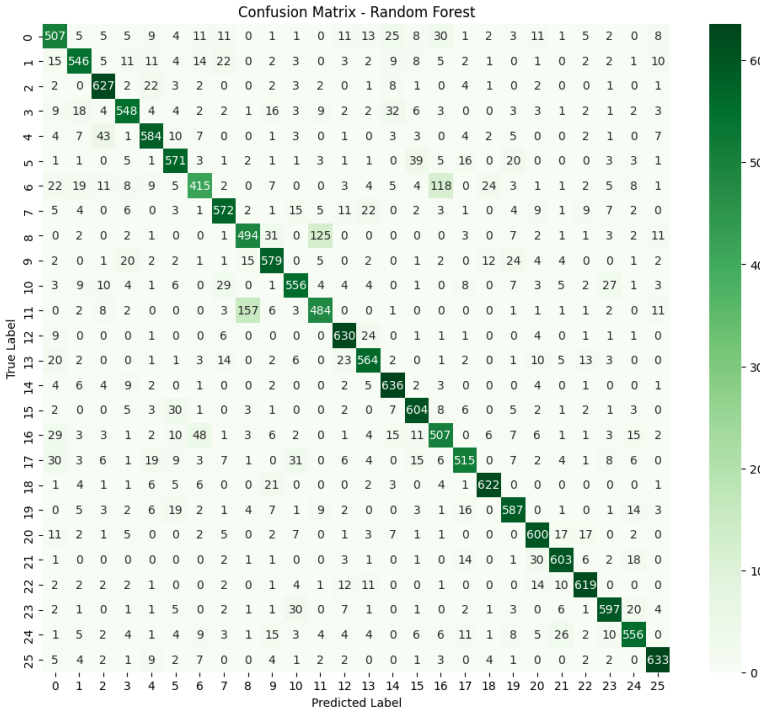
PCA & Decision Tree



HOG & Random Forests



PCA & Random Forests



Analysis, Discussion, and Future Work:

A. Analysis of Results

In this study, two feature extraction techniques, **Histogram of Oriented Gradients (HOG)** and **Principal Component Analysis (PCA)**, were employed to represent the EMNIST letter dataset. Subsequently, **Decision Tree (DT)** and **Random Forest (RF)** classifiers were applied, and their performances were evaluated using 5-fold cross-validation and test set metrics.

The key observations are summarized as follows:

1. **Decision Tree with HOG features**:
   - Cross-validation accuracies across folds were consistent, indicating stability in model training.
   - Test set accuracy reached approximately 64.54 % on the test data while by the validation data 65.91% , demonstrating that HOG features effectively capture the gradient and structural information of letters.
   - The confusion matrix revealed that letters with similar shapes (e.g., 'I' and 'L') and (e.g., 'G' and 'Q') were occasionally misclassified, suggesting that DT struggles with visually similar classes. “it appears well in confusion matrix”
2. **Decision Tree with PCA features**:
   - Dimensionality reduction via PCA significantly reduced the feature space, lowering computational cost.
   - Test accuracy was slightly lower than with HOG features **57.62%** on test data and on validation data **59.88%** ,implying that PCA, while efficient, may discard some discriminative information critical for letter distinction.
3. **Random Forest with HOG features**:
   - RF outperformed DT, achieving higher test accuracy **88.887%,** due to its ensemble nature, which reduces variance and mitigates overfitting.
   - The confusion matrix showed fewer misclassifications, particularly for visually similar letters, highlighting RF’s robustness against noise and feature ambiguities.
4. **Random Forest with PCA features**:
   - Accuracy was lower than HOG-based RF but still competitive **82%.**
   - PCA features enabled faster training times and reduced memory usage, although some fine-grained information essential for classification was lost.

These results indicate that **HOG features are more suitable for capturing letter-specific patterns**, while PCA provides a compact representation at the cost of some accuracy. Ensemble methods like Random Forest consistently outperform single Decision Trees by improving generalization.

5. **Random Forest with PCA and HOG features:**
    - After knowing that random forest is better than decision tree we use it with the PCA and HOG together to see the results
    - A Random Forest classifier was trained using combined HOG and PCA features, where HOG was used to extract discriminative shape features from images, and PCA was applied to reduce dimensionality before classification
    - The accuracy was slightly lower than HOG-based RF(**88.887%**) but still bigger than PCA-RF(**82%**) by **6%** accuracy = **88%.**

B. Discussion:

1. **Advantages**
    - **HOG Features**: Capture detailed structural information of letters, which enhances the classifier's ability to distinguish between visually similar classes.
    - **PCA**: Reduces feature dimensionality and computational load, making the model scalable to larger datasets.
    - **Random Forest**: Mitigates overfitting common in single Decision Trees and handles high-dimensional data effectively.
2. **Disadvantages**
    - **Decision Tree**: Susceptible to overfitting, particularly with high-dimensional HOG features, and sensitive to noise.
    - **PCA**: May remove subtle features necessary for distinguishing letters with minor variations.
    - **Random Forest**: Computationally expensive for very large datasets due to multiple trees and ensemble averaging.
3. **Algorithm Behavior**
    - The misclassification patterns observed in the confusion matrices align with expectations; letters with high visual similarity are prone to errors, which are consistent with the limitations of pixel-based or gradient-based representations.
    - Random Forests demonstrated superior performance due to their ensemble averaging mechanism, which reduces variance and improves robustness.

C. Future Work

Based on the current study, several future enhancements are suggested:

1. **Deep Learning Approaches:**
   - Implement Convolutional Neural Networks (CNNs), which can automatically extract hierarchical spatial features from images, potentially improving accuracy on visually similar letters.
2. **Data Augmentation:**
   - Apply rotations, translations, and scaling artificially increase dataset diversity, which can reduce overfitting and enhance generalization.
3. **Feature Fusion:**
   - Combine HOG and PCA features or explore other descriptors like Local Binary Patterns (LBP), to capture complementary information.
4. **Hyperparameter Tuning:**
   - Perform extensive tuning of tree depth, number of estimators, and PCA components to identify optimal model configurations.
5. **Dimensionality Reduction Alternatives:**
   - Explore t-SNE or UMAP for nonlinear feature reduction, which might preserve subtle differences between letters better than PCA.
6. **Real-Time Deployment:**
   - Investigate model deployment on embedded or edge devices for real-time handwritten letter recognition, considering computational efficiency and memory constraints.
7. **Conclusion:**

The study demonstrates that classical machine learning techniques, when combined with effective feature extraction, can achieve respectable performance on handwritten letter recognition tasks.

However, leveraging deep learning and advanced augmentation techniques presents a promising direction for future research

Project Summary:

- The EMNIST dataset was utilized in this project, consisting of grayscale images representing handwritten English alphabet characters.

- The system was developed using Python, with scikit-learn serving as the primary machine learning library.

- A unified workflow was designed to integrate data preprocessing, feature extraction, model training, evaluation, and visualization.

- Each image underwent a preprocessing pipeline that included resizing, reshaping, normalization, and conversion of pixel values into an appropriate numerical format prior to feature extraction.

- The dataset comprises 26 distinct classes, with each class corresponding to a single English letter.

- To ensure equitable learning across all classes, the dataset was carefully labeled and balanced.

- Rather than relying on a single train–test split, cross-validation was employed to obtain more robust and reliable performance estimates.

- This evaluation strategy helps minimize bias and provides a more accurate assessment of model generalization capability.

- Two feature engineering techniques were applied:
    - Histogram of Oriented Gradients (HOG) to capture structural, shape, and edge-based information.
    - Principal Component Analysis (PCA) to reduce dimensionality while retaining the most significant patterns in the data.

- A Decision Tree classifier was implemented as a baseline model due to its simplicity and interpretability.

- To enhance performance, a Random Forest classifier was subsequently employed, leveraging ensemble learning to combine multiple decision trees and mitigate overfitting.

- Experimental results demonstrated that the Random Forest model achieved superior accuracy compared to the Decision Tree, primarily due to reduced variance and improved generalization.

- Additionally, HOG-based features consistently outperformed PCA, as they better preserve spatial and edge information. Certain letters exhibited lower classification accuracy due to visual similarities in handwritten forms, a trend clearly reflected in the confusion matrices.

References & Links:

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